

Health Effects to Dietary Sodium Reduction

Technical review of the source-split CVD model and preliminary results for prioritized countries

Resolve to Save Lives (RTSL) and University of Washington

July 2026

Recap and today's plan

Recap: where we left off (July 9 call)

From the July 9 meeting and discussion:

- **Interventions** — six policy levers in the source-split model (public procurement, media, low-sodium salt substitutes, front-of-pack labelling, salt targets, fiscal); media is retained in the registry but excluded from reported scenarios.
- **Deliverables** — peer-reviewed paper, RTSL report, and interactive Shiny tool.

Today we present the **current build** — updated since July 9 — and **preliminary results** for the prioritized countries.

Recap: what the July 9 discussion decided

Scope & data

- Sodium sources: **combined approach** — use country data where available, else the Bhat/Marklund GDP regression (Bhat et al. 2020). Drivers extend beyond GDP (condiments/bouillon in Viet Nam, Nigeria).
- Updated sodium source shares with new data. **Nigeria and Cameroon still missing (provisional)**
- Benchmark: compare the full package to the 100M-Lives 30% sodium cut (not a like-for-like validation, but a contextual cross-check).

Decisions locked

- **Fiscal policy**: now **INCLUDED** (provisional effect on the taxed packaged share); country-specific elasticity model is future work.
- **Media / behavior change**: **EXCLUDED** from reported scenarios (machinery retained in the registry).
- **LSS**: run scenarios **2/4/5**; potassium now **captured** via the SSaSS trial relative risks.
- **Public procurement**: 5% share assumed schools — being **revised down** for an adults-only model (RTSL to supply per-country shares).
- **Packaged food**: no longer constant — a **Euromonitor 2025–2030 trend** is now applied.
- **Ramp-up**: confirmed **2026→2030** linear, applied as end-state at full scale-up.

Today's agenda

1. **What's changed since the last presentation** — five updates: fiscal now *included* (provisional), media *excluded*, **LSS scenarios 2/4/5 with potassium** via the SSaSS trial RRs, a **Euromonitor packaged-food trend**, and an **adults-only revision** of the public-procurement share.
2. **Model inputs** — updated sodium source shares by country (now incl. Nigeria/Cameroon) and the applied salt-effect η for each intervention and the full package.
3. **Methods** — the source-split logic (sources $\rightarrow \eta$), the state-transition engine, and the current low-sodium salt substitute (LSS) model.
4. **Preliminary results** — baseline burden, projected trajectories by scenario, and deaths delayed across the 10 prioritized countries.
5. **Limitations & discussion** — remaining open items and the timeline to the August reinvestment deliverable.

Main changes since the last presentation

1. Fiscal policy: now included (provisional)

- **Was:** excluded pending a team decision (applied $\eta = 0$).
- **Now:** included with a provisional effect — a **10%** reduction on the *taxed portion* of packaged sodium $\times$ an assumed **100%** taxable-packaged share $\Rightarrow$ applied $\eta = 10\%$ on packaged.
- **Future refinement (not yet implemented):** a country-specific price-elasticity model (Green et al. 2013), stratified by country income level.

Provisional: both the effect (10%) and the taxable-packaged share (100%) are RTSL-to-confirm.

Open question

What effect size should we assume for taxes on high-sodium “junk foods”? The BMJ paper does not directly estimate price elasticities for these products. Sugary foods are the closest available category, but it is unclear whether those estimates can reasonably be extrapolated to salty foods.

2. Behavior change / mass media: excluded

- **Was:** a reported lever (its estimate had been revised toward a UK basis).
- **Now: excluded from all reported scenarios** (July 14 decision) — RTSL funds policy change, not mass media for behavior change.
- **Retained** in the registry and effect-size tables ($\eta = 1.0\%$ to discretionary *and* packaged) so it can be re-enabled later without re-plumbing.

3. LSS: scenarios 2/4/5 with potassium (SSaSS trial RRs)

- **Was:** a single hypertensive-only scenario, sodium arm only — framed as “no potassium channel / conservative lower bound”.
- **Now:** the **SSaSS trial relative risks** capture the potassium benefit — stroke incidence RR **0.90** *and* a stroke case-fatality multiplier **0.856** $\Rightarrow$ **case fatality is no longer fixed for LSS** (total-stroke RR 0.86 is not applied separately).
- **Three scenarios:** 2 whole-population (50%), 4 diagnosed HTN (33%), 5 treated HTN (25%) — coverage **provisional**; household uptake for 4/5 under investigation.

4. Packaged food: Euromonitor trend (no longer constant)

- **Was:** the packaged share was assumed **constant** over time.
- **Now:** a **Euromonitor 2025–2030** predicted trend (median APC $\approx 2.6\%/yr$, range 0.1%–6.2%) grows the packaged share to a 2030 end-state, **held flat after 2030**.
- **Composition shift only** — total sodium intake is unchanged; the other sources renormalise. Source: Euromonitor (Marklund).

Open question

Should the projected growth in packaged foods affect the baseline CVD burden? Holding total sodium intake constant makes this a composition shift only. Allowing it to increase sodium intake—and therefore hypertension and age-specific CVD—would require considering why other evolving risk factors, such as obesity and air pollution, are not also modeled.

5. Public procurement: share under revision (adults-only)

- **Was:** a flat **5.0%** public share, which assumed **school coverage**.
- **Now:** the model is **adults-only**, so that share is being revised *downward*; RTSL owes the per-country adults-only values.
- **This run** used the stale 5.0% (code ran with a loud caveat) $\Rightarrow$ public-procurement and full-package results are **provisional**.

Model inputs

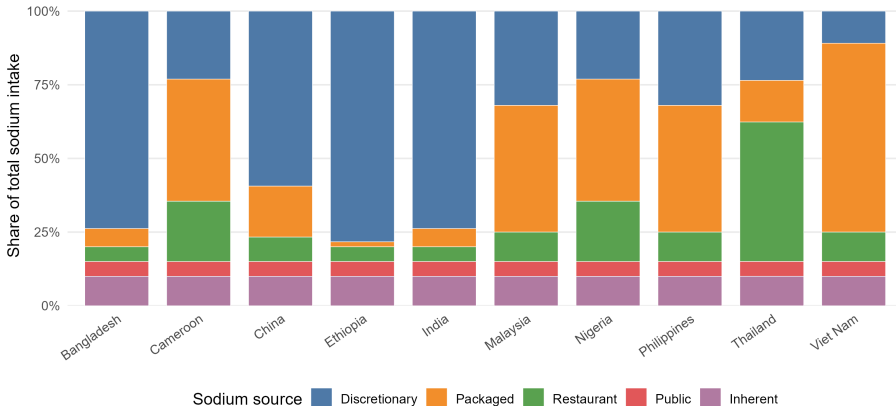
Model inputs at a glance

Input	Role
Baseline transition rates (GBD)	Age $\times$ sex $\times$ year $\times$ cause incidence (IR), case fatality (CF), background & COVID mortality, population (N_x).
BP distribution	Mean/SD systolic BP by age $\times$ sex, raised-BP fraction, diabetes prevalence.
Baseline sodium intake	2024 mean intake, split into five sources.
Sodium source shares	Five-source split of baseline sodium per country (this section).
Packaged-food trend	New: Euromonitor 2025–2030 predicted trend (median APC $\approx$ 2.6%/yr) grows the packaged <i>share</i> to a 2030 end-state, held flat after — composition shift only, total intake unchanged.
GBD relative risks	Log-linear RR per 10 mmHg SBP; sets bin-specific baseline incidence.
Ettehad relative risks	RR per 10 mmHg SBP reduction by BP bin $\times$ cause.
Scale-up schedule	Linear ramp 2026→2030, then held.

The source-split logic and the packaged-food trend are the changes from the earlier single-pool build; everything downstream of η is unchanged.

Sodium source shares by country

Baseline sodium is partitioned into **five mutually exclusive sources** (shares sum to 1). *inherent* sodium is **never targeted**; each policy acts only on the source(s) it can reach.



Applied salt-effect (η) by intervention

Each intervention has a documented **relative reduction** applied to its **target source**. Values match `code/07_run_interventions.R` exactly.

Intervention	Source	Applied reduction	Note
Public food procurement & service	public	20.0%	
Behaviour change / media campaigns	discretionary	1.0%	retained in registry; excluded from reported scenarios
Behaviour change / media campaigns	packaged	1.0%	retained in registry; excluded from reported scenarios
Front-of-pack labelling (FOPL)	packaged	13.8%	
Salt targets (reformulation)	packaged	20.0%	
Fiscal policy (salt tax)	packaged	10.0%	
Low-sodium salt substitutes (LSS)	discretionary	15.0%	sodium->SBP SENSITIVITY only; base case uses SSaSS trial RRs

Media (retained in the registry, excluded from reported scenarios) acts on discretionary **and** packaged (1.0% each). FOPL, salt targets and fiscal all act on **packaged** and compose multiplicatively when combined: e.g. FOPL (13.8%) + salt targets (20%)
 $\Rightarrow 1 - (1 - 0.138)(1 - 0.20) = \mathbf{31.0\%}$. **Fiscal is now *included* (provisional): 10% on packaged = a 10% effect on the assumed 100% taxed packaged share (RTSL to confirm).**

Methods

Methods

Low-sodium salt substitute (LSS) model — current version

Trial-RR decomposition (no double-count)

Stroke incidence RR $**0.90**$ (ischemic + hemorrhagic) *and* a conditional case-fatality multiplier $**0.77/0.90 = 0.856**$ on stroke CF (Neal et al. 2021). IHD/HHD unaffected $\Rightarrow$ **case fatality is no longer held at baseline for LSS.**

- **Three scenarios (coverage provisional, RTSL/HEARTS to confirm):** **sc. 2** whole-population discretionary (adjustable coverage 50%); **sc. 4** diagnosed HTN (33%); **sc. 5** treated HTN (25%). **Household uptake for 4/5 is under investigation.**
- **Sodium $\rightarrow$ SBP path retained as a documented sensitivity only** (the 15%-of-discretionary value, sensitivity 0.12–0.18); never combined with the trial-RR path.

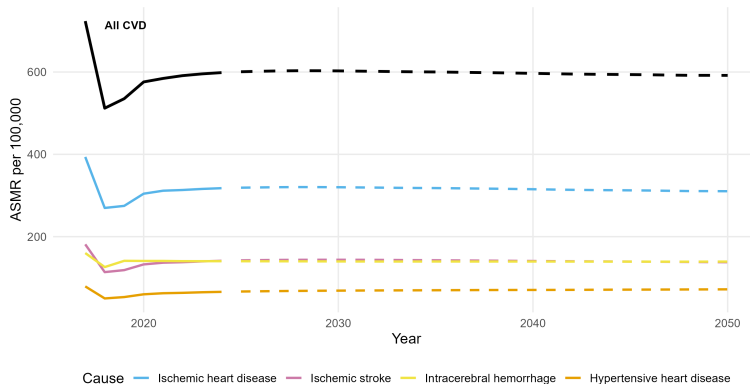
SSaSS context: 75% NaCl / 25% KCl substitute, ITT -350 mg/day sodium vs. a 4.3 g/day baseline, observed -3.34 mmHg SBP (much of it potassium-driven, +803 mg/day) (Huang et al. 2026).

Open question

Which approach should we use? Either model population-wide sodium and potassium changes through SBP, following Marklund, or restrict the SSaSS trial RRs to high-risk adults with hypertension—including undiagnosed, diagnosed untreated, and treated individuals. Applying SSaSS effects to the entire population may not be justified.

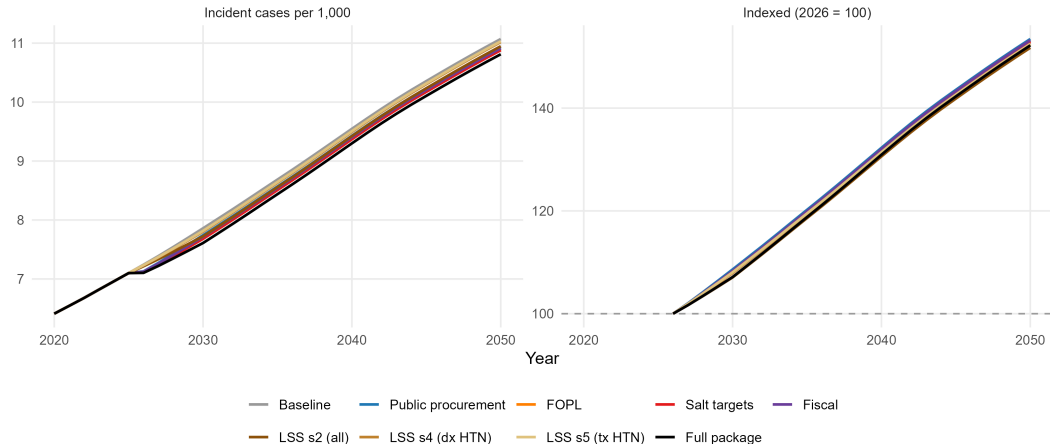
Baseline and projections

Baseline CVD burden, 2017–2050



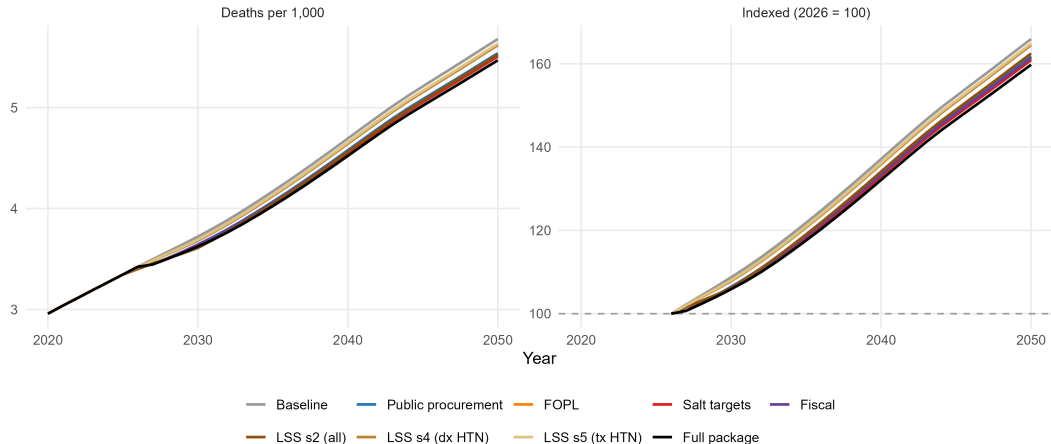
Age-standardised mortality (WHO standard, ages 20–95+) declines with forecasted background-mortality and case-fatality trends; absolute CVD deaths still rise with population growth and ageing. Solid = historical, dashed = projected.

Projected incidence by scenario



Incident CVD cases per 1,000 (left) and indexed to 2026 = 100 (right). Scenarios coincide until scale-up begins, then fan out; the **full package** falls fastest. **Fiscal now separates from baseline** (applied $\eta = 10\%$ on packaged, provisional).

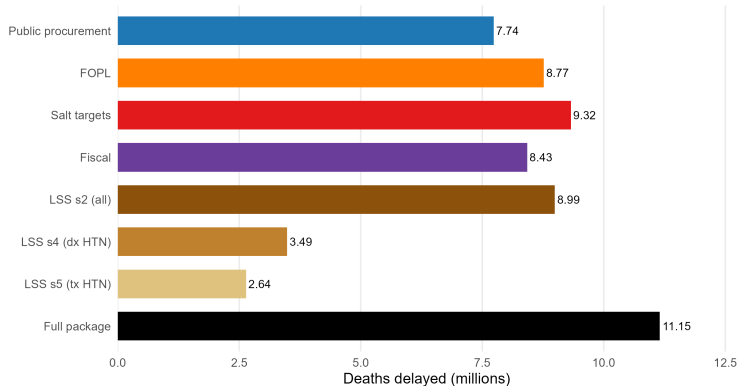
Projected deaths by scenario



CVD deaths per 1,000 (left) and indexed to 2026 = 100 (right). Because mortality responds to the **prior** year's incidence, the scenarios' divergence lags the incidence divergence by about a year.

Main results

Deaths delayed by scenario, 2026–2050



Headline

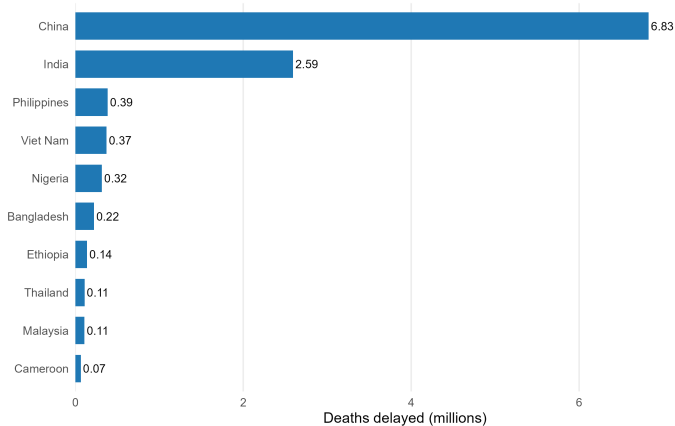
The **full package** delays **11.15 million** CVD deaths across 10 countries (3.2% of baseline CVD deaths in the window; ~446 thousand/year).

Deaths delayed by scenario — detail

Scenario	Deaths delayed (millions)	Avg per year (thousands)	% of baseline
Public food procurement & service	7.738	309.5	2.23%
Front-of-pack labelling (FOPL)	8.766	350.7	2.52%
Salt targets (reformulation)	9.324	372.9	2.68%
Fiscal policy (salt tax)	8.425	337.0	2.42%
LSS sc. 2 (whole-population)	8.993	359.7	2.59%
LSS sc. 4 (diagnosed HTN)	3.488	139.5	1.00%
LSS sc. 5 (treated HTN)	2.642	105.7	0.76%
Full package	11.150	446.0	3.21%

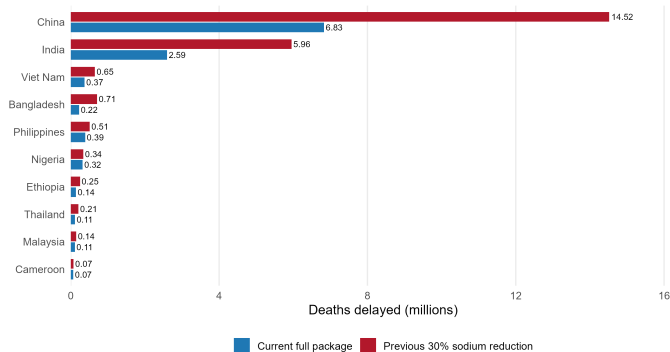
Among the reported policies, **salt targets** delays the most (9.32 M), then **FOPL** (8.77 M), **public procurement** (7.74 M, provisional), and **fiscal** (8.43 M, provisional). **LSS (SSaSS trial-RR)** runs as three scenarios: whole-population **sc. 2** 8.99 M, diagnosed-HTN **sc. 4** 3.49 M, treated-HTN **sc. 5** 2.64 M — targeting discretionary salt, which dominates intake in the most populous countries. Media is excluded from reported scenarios.

Full package: deaths delayed by country



Absolute counts are dominated by the most populous countries (China, India); per-capita impact depends on baseline burden and local source structure.

Benchmark: full package vs. the 100M-Lives 30% sodium cut



Contextual cross-check

Across the 10 selected countries, the current **full package** delays **11.15 M** CVD deaths vs. **23.36 M** for the previous **100 Million Lives Saved** flat 30% population sodium reduction ($\approx 0.48\times$). The package delivering less is *expected*: it applies **source-limited** policy effects, not a uniform 30% intake cut.

Limitations

Limitations and interpretation

- **Deaths delayed, not averted (sodium→SBP policies).** Case fatality is held at baseline for public procurement, FOPL, salt targets and fiscal, so mortality is shifted later; totals depend on the accounting window. **LSS is the exception** — its trial-RR pathway lowers stroke case fatality, so LSS *averts* some deaths.
- **Point estimates only.** The per-source reductions are deterministic — no probabilistic sensitivity analysis yet.
- **LSS coverage/uptake provisional.** The trial-RR pathway captures the potassium benefit (stroke RR 0.90; stroke CF $\times 0.856$), so LSS is *no longer* a no-potassium lower bound; the open uncertainty is provisional coverage (sc. 2/4/5) and household uptake for scenarios 4/5.
- **Fiscal policy included provisionally** — a fixed 10% effect on the assumed 100% taxed packaged share (applied $\eta = 10\%$ on packaged); the country-specific price-elasticity model (Green et al. 2013) is not yet implemented.
- **Provisional source shares.** Nigeria/Cameroon values are working estimates; several others rest on single-study or older survey data.
- **Heterogeneous observation periods.** Documented effects span 1–20 year windows and are not yet harmonized onto a common footing.
- **Independent-cause structure.** The four causes share one population but are modelled independently.

Discussion and next steps

Pending decisions — for discussion

Decided since the last meeting (no longer open)

Fiscal — *in* (provisional $\eta = 10\%$ on packaged). **Media** — *out* of reported scenarios (retained in the registry).

LSS — scenarios 2/4/5, potassium captured via the SSaSS trial RRs.

Still open

1. **Taxable-packaged-share assumption** — fiscal currently assumes 100% of packaged is taxed (likely too high); confirm, then move to the income-stratified elasticity model (Green et al. 2013).
2. **Adults-only procurement share** — RTSL to deliver per-country public shares to replace the stale school-inclusive 5.0%.
3. **LSS coverage & household uptake** — confirm scenario-2/4/5 coverage; household uptake/spillover for scenarios 4/5 still under investigation.
4. **Observation-period harmonization** — effects measured over 1–20 yr. End-state at full scale-up, linear over the observed window, or a country-tier heuristic?
5. **Source-of-sodium strategy** — how hard to push for country-specific shares vs. the Bhat/Marklund GDP regression (Bhat et al. 2020)? Confirm the Euromonitor leaf-category selection for the packaged trend.
6. **Policy framing** — lives-saved maximization vs. a prevention/counterfactual framing (e.g. Ethiopia).

Next steps and timeline

- **Inputs partly locked** — fiscal *in* (provisional), media *out*, LSS method set (SSaSS trial RRs, scenarios 2/4/5), ramp-up 2026\$→\$2030 confirmed.
- **Still pending** — RTSL's adults-only procurement shares, confirmation of the Euromonitor packaged-trend categories, and the fiscal price-elasticity refinement (Green et al. 2013).
- **Add the economic layer** — policy costing plus benefit–cost outputs (NPV, BCR) and DALYs.
- **Primary deliverable** — the **BP reinvestment proposal** memo (tables + figures), target **August 2026**.
- **Paper** is secondary; coordinate with Matti on the Bhat/Marklund inputs to avoid scooping the forthcoming paper.

Discussion

With fiscal/media/LSS now decided, where should we push hardest before the August reinvestment deliverable — the pending data (adults-only procurement shares, packaged-trend categories), the fiscal elasticity refinement, or the economic layer?

Appendix

Appendix: per-intervention methods

The following slides document each intervention's **evidence base** and the **applied** η used in the model. Applied values are produced by `reports/report.RMD` and `mirror code/07_run_interventions.R`; study effect sizes are cited from source.

Also in this appendix: documented source-share values (Table M1) and the per-scenario η decomposition by country.

A. Public food procurement

Target source: public (institutional meals). **Applied η :** 20% of the public source.

- **Evidence:** Gase et al. (2011) — modeling study, LA County. Weighted **-233 mg/day** ($\approx 7\% \approx \$20.6\%$ of public-meal sodium).
- **Source adjustment:** for a country with share x of sodium from institutional meals, population impact $= 0.20 \times x$. Worked: Viet Nam $\$ = 0.20 \times 5.0\% = \$ 0.0100$.
- **Share UNDER REVISION (provisional):** the flat 5.0% public share assumed **school coverage**; this model is **adults-only**, so the share is being revised *downward*. This run used the stale 5.0% (code ran with `ADULT_POPULATION_ONLY = FALSE`, printing a loud caveat); RTSL owes the adults-only per-country shares. Public-procurement and full-package results are therefore **provisional**.
- **Caveats:** modeled (not observed); US-specific setting; small affected population share.

B. Behavior change / mass media

Target source: discretionary + packaged. **Applied η : 1.0% to each.**

- **Evidence:** Do et al. (2016) — COMBI Vietnam, **5%** fall in mean sodium after a 1-year multi-component campaign. Collins et al. (2014) — UK economic evaluation, **2%** [1–5%].
- The estimate was revised toward a UK basis; a conservative 1.0% is applied to each of discretionary and packaged.

Open question

RTSL funds **policy change, not mass media for behavior change** (Laura). Keep this lever in the model, or drop it?

C. Low-sodium salt substitutes (LSS)

Base case: the **SSaSS trial relative risks** (approach B), which capture the potassium benefit — *not* the sodium-arm-only path.

- **Evidence:** Neal et al. (2021) (SSaSS) — -350 mg/day ITT sodium reduction vs. a 4.3 g/day baseline; observed -3.34 mmHg SBP, much of it potassium-driven (+803 mg/day).
- **Applied RRs:** stroke incidence RR **0.90** (ischemic + hemorrhagic) and a conditional stroke case-fatality multiplier **0.856** ($= 0.77/0.90$); total-stroke RR **0.86** is not applied separately. IHD/HHD unaffected; **stroke CF no longer held at baseline.**
- **Scenarios** (Huang et al. 2026): 2 (whole-population, coverage 50%), 4 (diagnosed HTN, 33%), 5 (treated HTN, 25%) — *coverage provisional*; household uptake for 4/5 under investigation.
- **Sensitivity only:** the sodium→SBP path (15% of discretionary, 0.12–0.18); never combined with the trial RRs.

D. Front-of-pack labeling

Target source: packaged (estimate already from packaged-food purchases). **Applied η :** 13.8% of packaged.

- **Evidence:** Taillie et al. (2024) — Chile's mandatory warning-label law (observed). **-73.3 mg/capita/day (-13.8%)** in sodium purchases (2,844 households, 2013–2019).
- **Caveat:** purchases-to-intake translation is uncertain.
- **Alternative:** Mexico found $\approx -7\%$ in packaged sales (≈ -38 mg/capita/day).

E. Mandatory salt targets (reformulation)

Target source: packaged. **Applied η :** **20% of packaged** (conservative; 15% floor for sensitivity).

- **Policy:** South African mandatory maximum sodium limits (2013; targets implemented 2016 and 2019).
- **Evidence:** Strauss-Kruger et al. (2022) — African-PREDICT, **−15.2%** sodium; baseline/follow-up straddle the target dates, so likely an **underestimate**.
- **Alternatives:** (Gaziano et al. 2025) −8.4% (rural, spot urine); (Charlton et al. 2021) −18.4%; UK (He, Pombo-Rodrigues, and MacGregor 2013; Song et al. 2023); Mexico modeling (Muciño-Sandoval, Barrientos-Gutiérrez, et al. 2025).

F. Fiscal policy / taxation

Target source: packaged. **Applied η :** 10% of packaged — now *included* (provisional).

- **Evidence:** Saxena et al. (2025) — Philippines, modeled 20% tax on foods high in salt/sugar; **−100 mg/day (−10%)** at year 20.
- **Current MVP (provisional):** a fixed **10%** effect on the **taxed portion** of packaged sodium, times an assumed **100%** taxable-packaged share $\Rightarrow$ applied $\eta = 10\%$ on packaged. *Both factors are RTSL-to-confirm assumptions*; a 100% taxable share is likely too high.
- **Planned refinement (not yet implemented):** a country-specific price-elasticity model (Green et al. 2013), stratified by country income level, replacing the fixed effect.
- **Alternatives:** India (Roche et al. 2026); Australia (Davies et al. 2026; Cobiac et al. 2017); systematic review (Dodd et al. 2020) (1–11% reduction).

Appendix: sodium source shares — documented values

Country	Discretionary	Packaged	Restaurant	Public	Inherent
Viet Nam	0.109	0.641	0.100	0.050	0.100
Philippines	0.320	0.430	0.100	0.050	0.100
Bangladesh	0.738	0.062	0.050	0.050	0.100
China	0.594	0.173	0.083	0.050	0.100
Ethiopia	0.783	0.017	0.050	0.050	0.100
India	0.738	0.062	0.050	0.050	0.100
Malaysia	0.320	0.430	0.100	0.050	0.100
Thailand	0.235	0.141	0.473	0.050	0.100
Nigeria	0.231	0.415	0.205	0.050	0.100
Cameroon	0.231	0.415	0.205	0.050	0.100
default	0.430	0.279	0.142	0.050	0.100

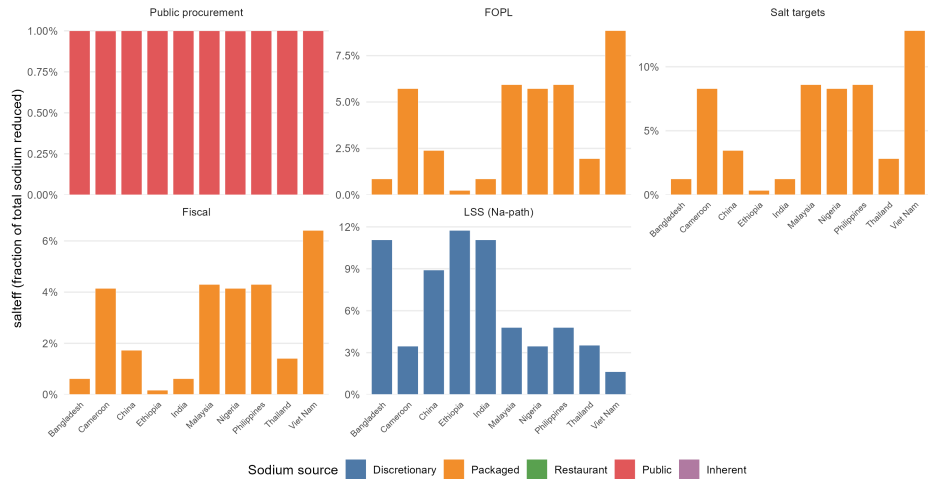
Fractions of total intake, renormalised to 1.0. The default row (mean of the ten) applies to any country without its own row.

Appendix: packaged-food trend (Euromonitor)

The packaged *share* follows a Euromonitor 2025–2030 predicted trend (median APC per year, over non-overlapping sodium-relevant leaf categories), applied as the 2030 end-state and **held flat after 2030**. Total intake is unchanged — only *where* sodium comes from shifts. Median APC $\approx$ 2.6%/yr (range 0.1%–6.2%).

Country	APC (%/yr)	Packaged share 2024	Packaged share 2030
Bangladesh	6.18	6.2%	8.2%
India	5.87	6.2%	8.1%
Nigeria	4.23	41.5%	46.6%
Philippines	3.27	43.0%	47.0%
Thailand	2.82	14.1%	15.9%
Viet Nam	2.33	64.1%	66.7%
Malaysia	1.56	43.0%	44.9%
China	1.42	17.3%	18.3%
Ethiopia	0.70	1.7%	1.8%
Cameroon	0.06	41.5%	41.5%

Appendix: η decomposition by scenario and country



Per-scenario `salteff` for every country, stacked by source contribution (note free y-axes). Fiscal now contributes a 10% reduction of the packaged source (provisional).

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